

Technical Datasheet WIN-6NTC-Modbus

The Master Display Unit polls distributed Sensor Nodes via RS485 and acts as a central gateway to upstream PLC/SCADA systems. The unit is sturdy, built for continuous 24/7 operation, and exceptionally easy to use.

Master Display Module: -



The modules are housed in an industrial casing featuring an integrated touchscreen display, exposed connectors, and an onboard buzzer. Soft settings for Wi-Fi, Modbus parameters, individual sensor naming, and custom thresholds are managed entirely through the direct display interface or the built-in web portal.

The design of the modules incorporates 'resettable Fuses' to safeguard against reverse polarity connection both for Power and Communication ports. In addition, it features an active audio-visual alarm system with a front-panel mute button to immediately flag thermal breaches across all 24 monitoring zones. Users can connect up to 4 distributed Sensor Nodes to the Master Display Unit. Each node features 6 NTC channels, providing a total monitoring capacity of 24 independent sensors.

Specifications

General –

Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 - 24 VDC
Operating Temperature	-20°C to +80°C (-4°F to +176°F)
Storage temperature	-20°C to +80°C (-4°F to +176°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



Additional Features: -

All inputs and communication ports isolated

Input power reverse polarity safety

ESD Safety IEC 61000-4-2, ± 30 KV contact, ± 30 KV air

EFT IEC 61000-4-4, 50A (5/50ms)

750V isolation.

CRC Error check.

Operator Interface & Navigation Guide

- The Master Display Unit features a touchscreen interface designed for rapid diagnostics and seamless local configuration.

1. Initial Network Provisioning (Wi-Fi Setup):

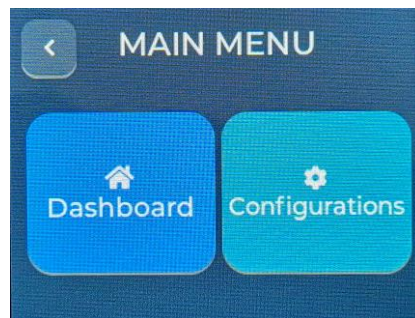
Upon first boot—or when a network change is required—the device enters Access Point (AP) mode. The screen displays a "WiFi Setup Required" prompt, broadcasting its own local network (Busbar Monitor). Users simply connect to this network via a mobile device or laptop using the password 12345678 and navigate to the provided IP address (192.168.4.1) to securely pass local Wi-Fi credentials to the unit.



2. Home Screen & Main Navigation

Once provisioned, the device boots to the Home Screen, displaying the company logo, current date/time, and system title. Tapping the [>] icon in the top right corner opens the **Main Menu**, which splits the interface into two primary pathways:

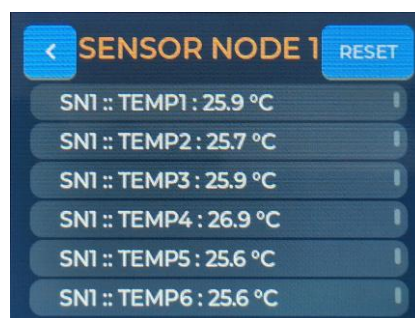
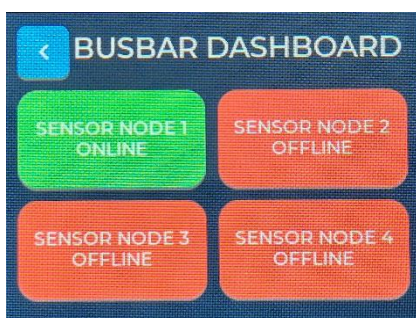
- **Dashboard:** For live telemetry and alarm management.
- **Configurations:** For core system and network parameters.



3. Live Telemetry Dashboard

The Dashboard provides an immediate, color-coded health summary of the entire Modbus RS485 network.

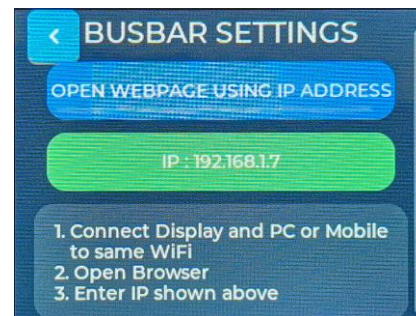
- **Node Overview:** Up to four connected Sensor Nodes are displayed as individual blocks. Active, healthy nodes highlight in **Green (Online)**, while disconnected or powered-down nodes highlight in **Red (Offline)**.
- **Sensor Drill-Down:** Tapping an Online node opens its specific 6-channel NTC data screen. This displays the live temperature for all 6 individual sensors (which can be assigned custom alphanumeric names via the web portal).
- **Alarm Management:** If a specific sensor breaches its programmed threshold, the display will flash to indicate the fault location. The operator can press the [**RESET**] button in the top right corner to acknowledge the fault and mute the audible buzzer for a 10-minute diagnostic window.



4. System Configurations Menu

Navigating to the Configurations menu grants operators access to three sub-menus for local system management:

- **Modbus Settings:** Allows administrators to Save the upstream communication parameters (i.e.: Slave ID and Baud Rate) directly from the screen or reset to factory defaults.
- **Busbar Settings:** Displays the unit's active local IP address (e.g., 192.168.1.7) alongside simple, on-screen instructions for accessing the advanced Web Configuration Portal via a PC or mobile browser.
- **Wi-Fi Settings:** Displays the currently connected network SSID. Pressing [**CHANGE WIFI**] forces the device to reboot back into the initial AP Setup mode, allowing the unit to be moved to a new facility or network without requiring a hardwired PC connection.



Web Portal Configuration Guide (Busbar Settings)

1. Accessing the WiFi Manager

- Navigating to the device's IP address opens the main WIFI Manager portal.
- Click the **[Configure WiFi]** button to proceed to the network selection page.

WiFiManager

Busbar_Monitor

Configure WiFi

Info

Exit

Update

No AP set

2. Selecting the Network

- The setup page displays a list of available Wi-Fi networks in range.
- Tap the desired network from the list to automatically populate the **SSID input field** or type the SSID manually.
- Enter the network security key into the **Password input field**.
- Click the **Save button** to commit the credentials.
- The device will then process the information and connect to the assigned Wi-Fi network.

WittelB-SN-ATH-SN5	 
WittelB-SN-ATH-SN2	 
AirFiber-DIqCcP	 
DIRECT-FE-HP Smart Tank 750	 
WittelB	 
WittelB-SN-ATH-SN1	 
WIN-SLD	 
WittelB-SN-TH-SN7	 

SSID

Password

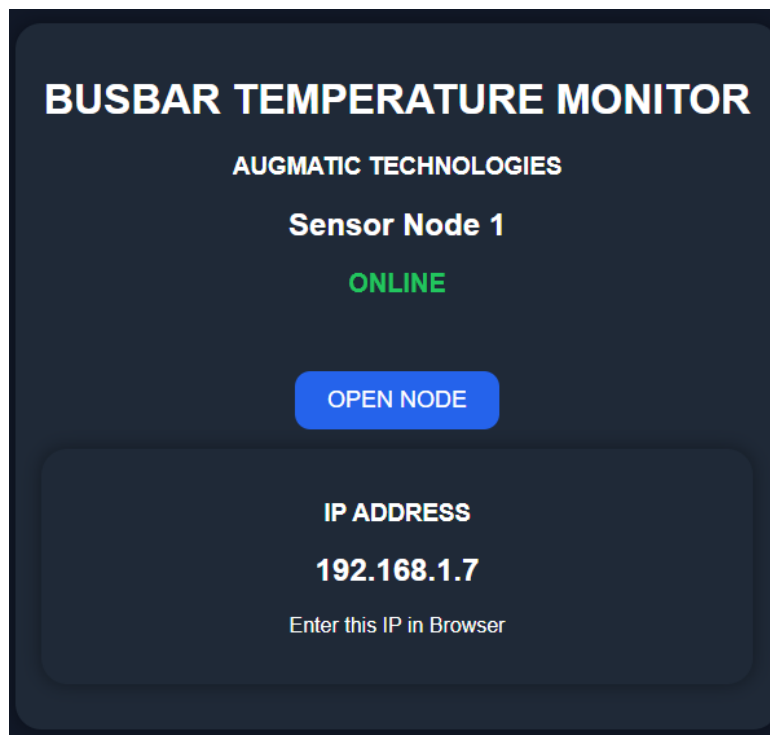
☐ Show Password

Web Portal Configuration Guide (Busbar Settings)

- The embedded web portal is dedicated to managing busbar temperature parameters, specifically sensor naming and thermal thresholds.

1. Accessing the Settings

- Enter the IP address shown on the physical display into a web browser.
- The home screen displays the current status of the sensor node. Click the [**OPEN NODE**] button to access its configuration page.



2. Customizing Sensor Names

- The interface displays the Current Name for each of the 6 NTC channels. To assign a custom identifier, enter the desired text into the corresponding New Name field.

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SENSOR NODE 1 SETTINGS

Current Name	New Name	Current Threshold	New Threshold	Reset
TEMP1	NTC1	100.0	New Threshold	RESET
TEMP2	New Name	100.0	105	RESET
TEMP3	R3	100.0	New Threshold	RESET
TEMP4	New Name	100.0	New Threshold	RESET
TEMP5	New Name	100.0	New Threshold	RESET
TEMP6	New Name	100.0	New Threshold	RESET

SAVE SETTINGS

Instructions

1. Current Name shows existing sensor name.
2. Enter New Name only if you want to change it.
3. Default Threshold = 100 C.
4. Enter New Threshold if required.
5. RESET button restores threshold to 100 C.
6. Press SAVE SETTINGS to save changes.
7. Display updates automatically.

3. Defining Alarm Thresholds

- The interface displays the Current Threshold for each sensor, which defaults to 100.0°C.
- To assign a different thermal alarm limit, enter the specific value into the corresponding New Threshold field.

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SENSOR NODE 1 SETTINGS

Current Name	New Name	Current Threshold	New Threshold	Reset
TEMP1	NTC1	100.0	New Threshold	RESET
TEMP2	New Name	100.0	105	RESET
TEMP3	R3	100.0	New Threshold	RESET
TEMP4	New Name	100.0	New Threshold	RESET
TEMP5	New Name	100.0	New Threshold	RESET
TEMP6	New Name	100.0	New Threshold	RESET

SAVE SETTINGS

Instructions

1. Current Name shows existing sensor name.
2. Enter New Name only if you want to change it.
3. Default Threshold = 100 C.
4. Enter New Threshold if required.
5. RESET button restores threshold to 100 C.
6. Press SAVE SETTINGS to save changes.
7. Display updates automatically.

4. Saving & Resetting Changes

- Click the green [SAVE SETTINGS] button to apply the new names and thresholds. The changes will automatically update on the physical display.
- Click the red [RESET] button on a specific sensor's row to restore its threshold back to the default 100.0°C.

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SENSOR NODE 1 SETTINGS

Current Name	New Name	Current Threshold	New Threshold	Reset
NTC1	New Name	100.0	New Threshold	RESET
TEMP2	New Name	105.0	New Threshold	RESET
R3	New Name	100.0	New Threshold	RESET
TEMP4	New Name	100.0	New Threshold	RESET
TEMP5	New Name	100.0	New Threshold	RESET
TEMP6	New Name	100.0	New Threshold	RESET

[SAVE SETTINGS](#)

Instructions

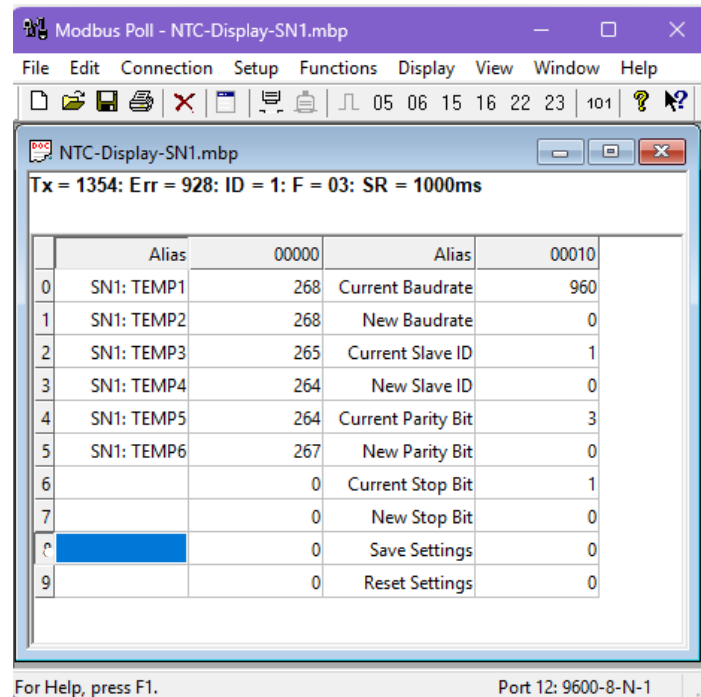
1. Current Name shows existing sensor name.
2. Enter New Name only if you want to change it.
3. Default Threshold = 100 C.
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MOSBUS Register Configuration

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	SN1: TEMP1	40001	0X03	RS485	16 Bit Unsigned int
2	SN1: TEMP2	40002	0X03	RS485	16 Bit Unsigned int
3	SN1: TEMP3	40003	0X03	RS485	16 Bit Unsigned int
4	SN1: TEMP4	40004	0X03	RS485	16 Bit Unsigned int
5	SN1: TEMP5	40005	0X03	RS485	16 Bit Unsigned int
6	SN1: TEMP6	40006	0X03	RS485	16 Bit Unsigned int
7	Display Baud Rate (Default:960)	40011	0x03	RS485	16 Bit Unsigned int
8	Enter New Baud Rate	40012	0x03	RS485	16 Bit Unsigned int
9	Display Slave ID (Default:1)	40013	0x03	RS485	16 Bit Unsigned int
10	Enter New Slave ID	40014	0x03	RS485	16 Bit Unsigned int
11	Display Parity (Default: None-3, Odd-1, Even-2)	40015	0x03	RS485	16 Bit Unsigned int
12	Enter New Parity	40016	0x03	RS485	16 Bit Unsigned int
13	Display Stop Bit (start-2/stop bit-1)	40017	0x03	RS485	16 Bit Unsigned int
14	Enter New Stop Bit	40018	0x03	RS485	16 Bit Unsigned int
15	Save Modbus Settings	40019	0x03	RS485	16 Bit Unsigned int
16	Reset Modbus Settings	40020	0x03	RS485	16 Bit Unsigned int

MOSBUS Mbpoll Configuration

- Here is an example of receiving data on Mb poll software.
- Open Mb-poll software connect using configuration setting (Default baud rate is 9600 and slave id is 1).
- After connecting you will get data as shown below



- As you can see in the image above, the resistor 40010 – 960 represents the present bund rate (9600) of the board and 40012 – 1 represents the present slave id of the board.

Write Single Register

Slave ID:

1

Send

Address:

11

Cancel

Value:

11520

Result

N/A

☐ Close dialog on "Response ok"

Use Function

☒ 06: Write single register
☐ 16: Write multiple registers

Write Single Register

Slave ID:

1

Send

Address:

13

Cancel

Value:

12

Result

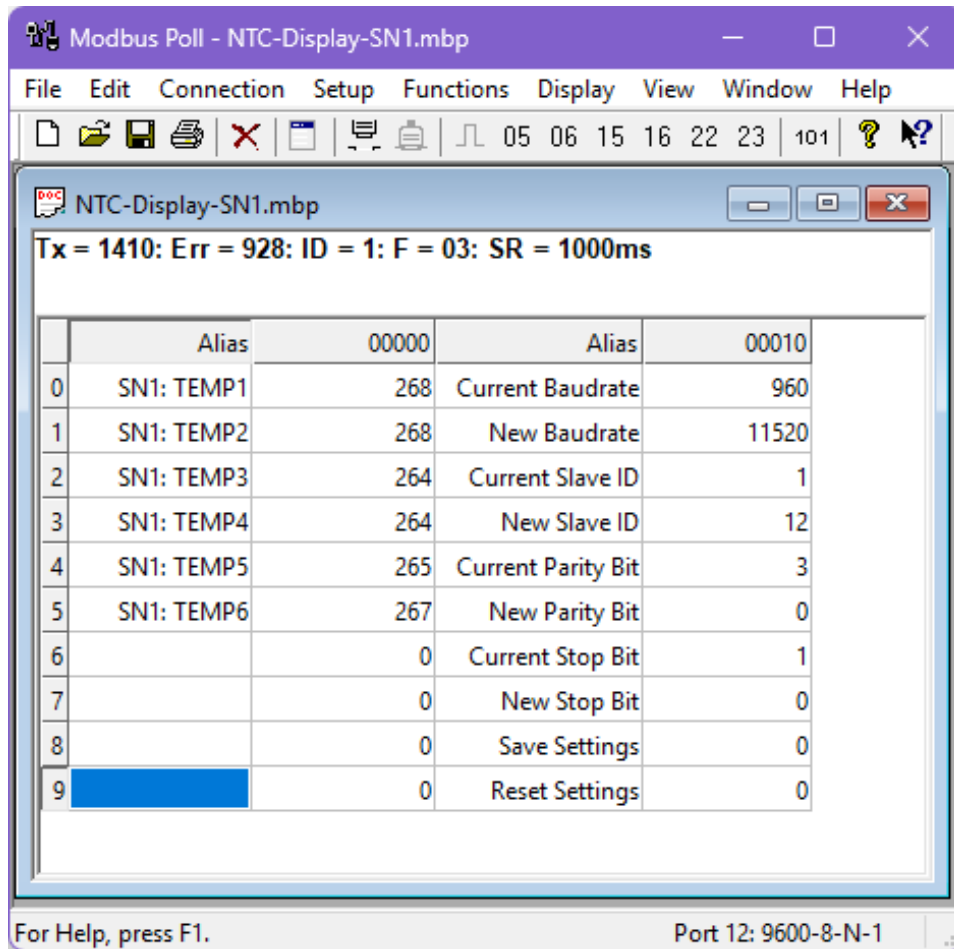
N/A

☐ Close dialog on "Response ok"

Use Function

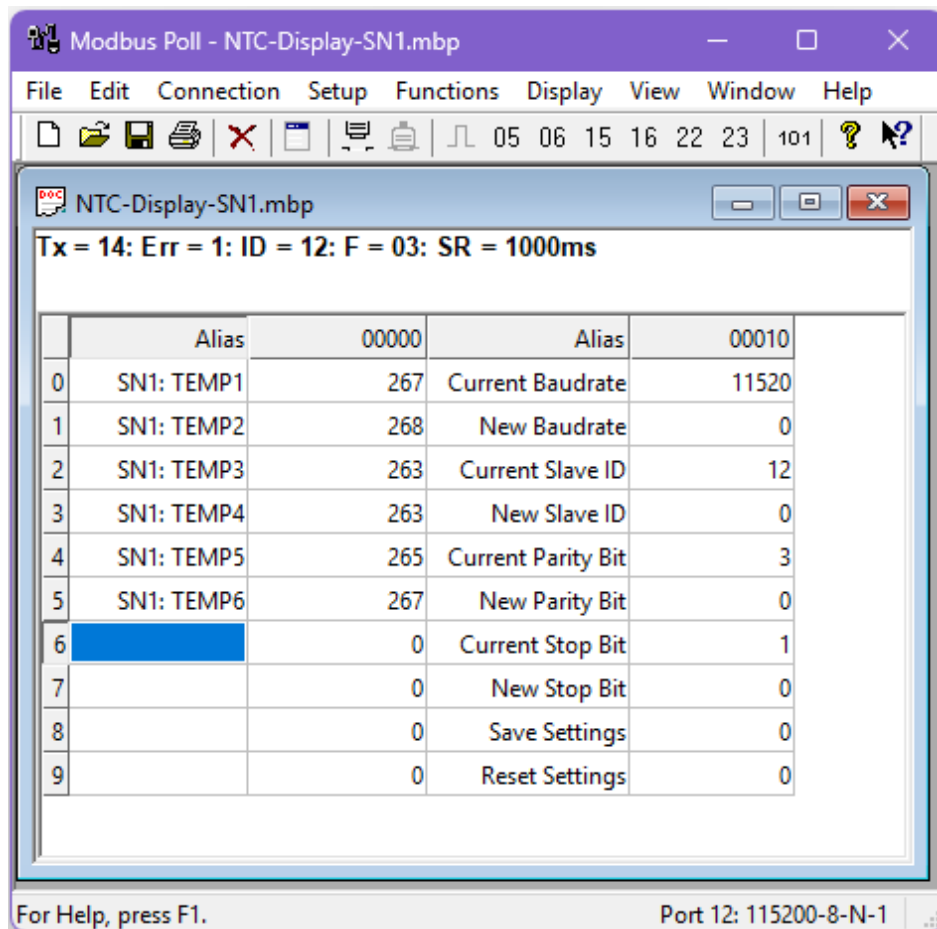
☒ 06: Write single register
☐ 16: Write multiple registers

- To change the baud rate and slave id you must enter a value on the 40011 resistors for the baud rate and 40013 for the slave id as shown in the image below.



- After that, press the SAVE button located in the Modbus Settings screen on the display, or remotely write a value of 1 to Modbus register 40019.
- If you want to enter the baud rate 9600/19200/38400/57600/115200etc. Enter it like this 960/1920/5760/11520etc.

- Then you will get the present value of 11520 at 40011 baud rate(115200) and 12 at 40013 slave id.



- Network Parameter Recovery: In the event of lost communication parameters, factory default settings (9600 Baud, Slave ID 1) can be instantly restored.
- Either press the RESET button within the touchscreen's Modbus Settings interface or remotely write a value of 1 to Modbus holding register 40020.

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